

Process

to differentiate

$x=1$
 $\text{int } a = \text{fork}();$

if ($a == 0$)
 $x = x + 1;$

$\text{pid} = \text{fork}();$
 $\text{print}(x);$

else

~~wait~~
 $\text{print}(x)$
 parent

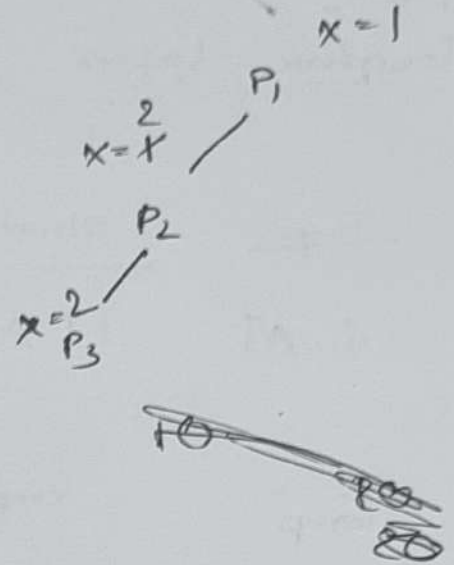
if ($\text{pid} == 0$)

// P_3 $\text{Print}(\text{GrandChild})$

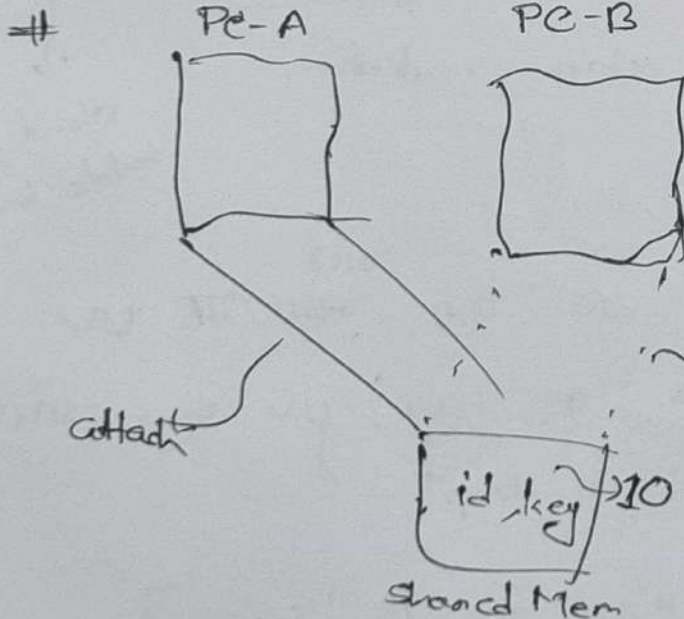
else

$\text{wait}();$
 // P_2

$\text{Print}(\text{child})$
~~parent~~



$\Rightarrow$ grand...
 child...
 par...



to read
 need to attach

overlapping

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Issue $\rightarrow$ Race Cond

ADALI

Adalimumab INN

(dirty Read type)

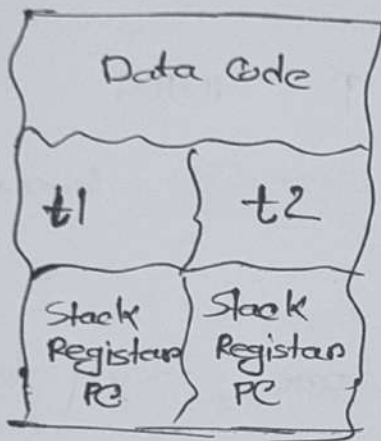
Threads

(Issues of Process)

Memory copy $\rightarrow$ Time consuming

Expensive

For more client not scalable



Stack $\rightarrow$ functions save
stack

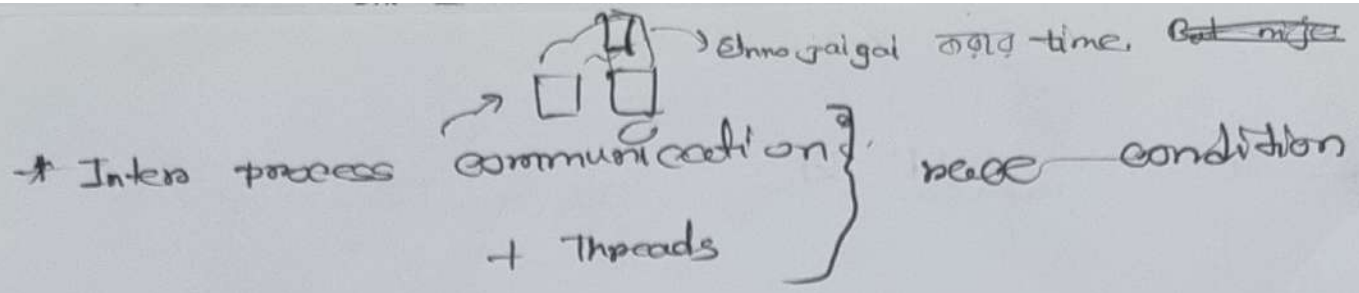
Stack $\rightarrow$ function, local
variable, return

save stack

Same data $\rightarrow$ different Queries $\rightarrow$ create Thread
diff data

Different data $\rightarrow$ different Queries $\rightarrow$ Create Process

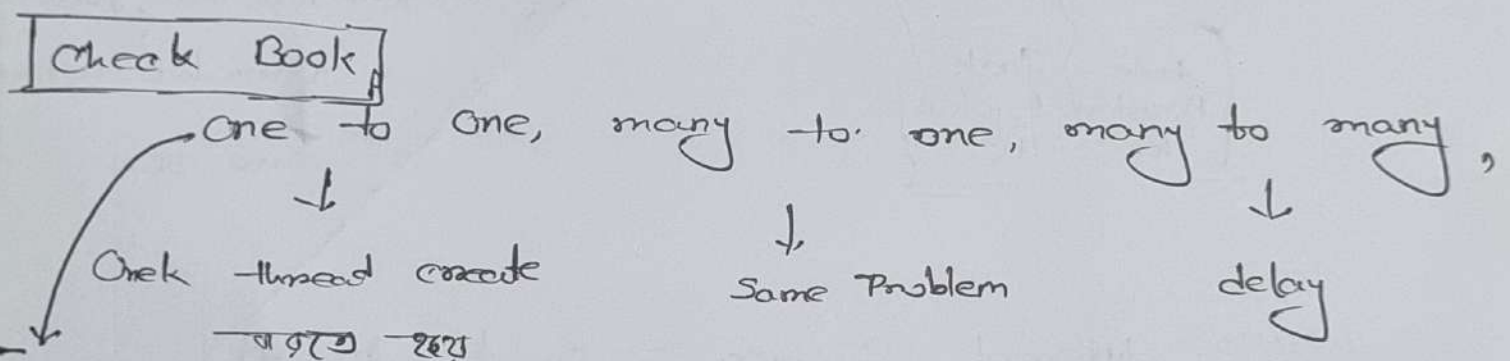
~~By default, process~~



[1, ..., 1M] data, even print

Multithreading Model

Users level thread → Error / Read / Write / Time ↑ / kernel level thread execution



Used in current level OS → need responsiveness, current machine has overheads

Two level model

↓

M2M, O2O

↓

mixed